

Product Requirements Document (PRD)

Project Name: Healthcare Triage & Management Application

Document Version: 1.0

This document outlines the product requirements, system specifications, architectural improvements, and a comprehensive feasibility assessment for a centralized healthcare platform connecting patients, doctors, and clinics.

1. Product Vision and Scope

The application aims to bridge the gap between emergency patient needs and clinical readiness by providing a unified ecosystem. It empowers patients with immediate remedial guidance and geolocation-based clinic discovery, equips doctors with AI-driven patient history retrieval, and prepares clinics for incoming emergencies through real-time alert systems.

2. Functional Requirements (Core Features)

2.1 Patient Clinical History Management

Description: A secure, centralized repository for all patient medical records, accessible by authorized healthcare providers and the patients themselves.

- Support for uploading and parsing medical documents (PDFs, lab results, imaging).
- Chronological timeline view of past diagnoses, prescriptions, and surgeries.
- Patient consent management module to grant or revoke access to specific clinics or doctors.

2.2 Remedial Measures and Triage

Description: An AI-driven symptom checker that provides immediate, evidence-based at-home care instructions while screening for critical emergencies.

- Symptom input interface (text or guided questionnaire).
- **Triage Engine:** Must prioritize red-flag symptoms and immediately route the user to emergency services if critical thresholds are met.
- **Remedial Output:** Delivery of standardized, safe, at-home remedial guidelines based on established medical protocols (incorporating strict liability disclaimers).

2.3 AI Chatbot for Medical Staff (History Search)

Description: A conversational interface enabling doctors to query a patient's comprehensive medical history rapidly.

- Natural Language Processing (NLP) to handle queries like, "Does this patient have a history of penicillin allergies?" or "Summarize the patient's cardiac events from 2024."
- Retrieval-Augmented Generation (RAG) architecture to ensure the bot strictly sources answers from the specific patient's uploaded EHR (Electronic Health Record), preventing medical hallucinations.
- Source citation linking directly to the original medical document.

2.4 Closest Clinic Suggestions

Description: A location-based recommendation engine for patients seeking immediate care.

- Integration with mapping APIs (e.g., Google Maps, Mapbox).
- Filtering by clinic specialty, current operating hours, and accepted insurance.
- Real-time estimated time of arrival (ETA) and routing instructions.

2.5 Clinic Registration and Dashboard

Description: A dedicated onboarding portal for clinics to join the network and manage their operational presence.

- Verification workflow requiring medical licenses and accreditation documents.
- Profile management (services offered, capacity, operating hours, staff roster).
- A centralized web-based dashboard for administrative staff to monitor incoming patients.

2.6 Pre-Arrival Emergency Alerts

Description: An early-warning system that notifies a selected clinic that an emergency patient is en route.

- Triggered automatically when a patient uses the "Closest Clinic" feature for an urgent issue, or manually by the patient/paramedic.
- Transmission of a critical data packet to the clinic (patient name, blood type, primary complaint, ETA).
- Multi-channel alerts (dashboard push notifications, SMS, and automated voice calls to the clinic's reception).

3. Proposed System Improvements & Additions

To ensure scalability, legal compliance, and market competitiveness, the following features should be integrated into the product roadmap:

Improvement Area	Recommended Feature	Justification
Security & Compliance	HIPAA / GDPR Compliance Framework	Healthcare data is highly regulated. End-to-end encryption, audit logging, and strict data residency protocols are mandatory, not optional.
Access Control	Role-Based Access Control (RBAC)	Nurses, receptionists, and doctors require different levels of access to patient data. A robust RBAC system prevents unauthorized data exposure.
Interoperability	HL7 / FHIR Standard Integration	To scale, the system must communicate with existing hospital systems. Adopting FHIR (Fast Healthcare Interoperability Resources) ensures data portability.
Patient Experience	Appointment Booking & Queue Management	Beyond emergencies, allowing users to book standard appointments and view live wait times adds daily utility to the app.
Remedial Safety	Human-in-the-Loop (HITL) Fallback	For ambiguous symptoms, the AI remedial measure system should seamlessly offer a telehealth consultation rather than guessing.

4. Feasibility Check

4.1 Technical Feasibility: Moderate to High Complexity

- **Data Architecture:** Storing and querying unstructured medical data (for the chatbot) requires a robust vector database and secure LLM integration. This is technically feasible but requires experienced AI/backend engineers.
- **Real-Time Infrastructure:** Emergency alerts require highly reliable, low-latency infrastructure (e.g., WebSockets, specialized push notification services). Uptime must be 99.99%.
- **Geolocation:** API integrations for location services are standard and highly feasible.

4.2 Legal and Regulatory Feasibility: High Risk

- **Medical Liability:** Providing "remedial measures" through software borders on providing medical advice. The application must be legally structured as a general information tool, not a diagnostic medical device (e.g., SaMD regulations).
- **Data Privacy:** Storing clinical history mandates strict adherence to healthcare data laws. Breaches carry severe financial and legal penalties.

4.3 Operational Feasibility: Moderate

- **Clinic Adoption:** The classic "chicken-and-egg" marketplace problem. Patients won't use the app if clinics aren't registered; clinics won't monitor the emergency dashboard if patients aren't using the app.
- **Workflow Integration:** Emergency rooms already have established triage protocols. The app's pre-arrival alert must integrate seamlessly into their current screens (or via loud auditory alerts) rather than forcing nurses to monitor a separate device.

4.4 Economic Feasibility: High Initial Investment Required

- **Development Costs:** Building secure medical databases, licensing medical-grade LLMs, and ensuring rigorous security compliance requires significant upfront capital.
- **Monetization Strategy:** The platform can be monetized via a SaaS model for clinics (charging for priority listing or the advanced CRM dashboard), while keeping the patient-facing app free to drive user adoption.